

temperatures 16°C above absolute zero. Thus, it can be utilised to create a superconducting qubit when arranged in a structure called a Josephson junction. Researchers investigated the impact of temperature on the crystal structures and electrical properties of NbNx thin films grown on AlN template substrates. They showed that the spacing of atoms in the two materials was compatible enough to produce flat layers. This interface between the AlN substrate, which has a wide bandgap, and NbNx is essential for future quantum devices, such as Josephson junctions. Superconducting layers that are only a few nanometers thick and have high crystallinity can be used as detectors of single photons or electrons.

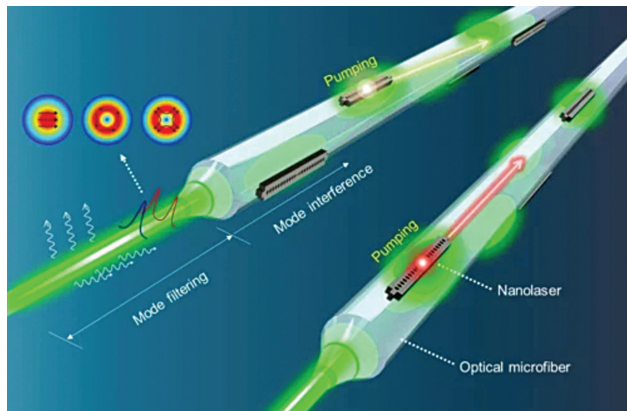
New technique found to improve free-space optical communication

New signal-processing algorithms have been shown to mitigate the impact of turbulence in free-space optical experiments, potentially bringing “free space” internet a step closer to reality. A team of researchers from Aston University’s Aston Institute of Photonic Technologies and Glasgow University used commercially available photonic lanterns, a commercial transponder, and a spatial light modulator to emulate turbulence. They simultaneously transmitted multiple data signals using different spatially shaped beams of light using a so-called photonic lantern. Turbulence changes the shape of the beams, often losing the signal if only a single simple shape is transmitted and detected, but by detecting light with these shapes using a second lantern, more of the light is collected at the receiver, and the original data can be unscrambled. This can greatly reduce the impact of the atmosphere on the quality of the data received in a technique known as Multiple-Input Multiple-Output (MIMO) digital signal processing. This project aims to provide the internet performance of a pure fibre connection without the need to install cables.

New optical approach can enable much faster communication

A new all-optical approach for driving multiple highly dense nano laser arrays has been developed by researchers in Korea. This method can enable chip based optical communication links that process and move data much faster than current electronic based devices. Researchers have demonstrated that densely integrated nano laser arrays—in which the lasers are just 18 microns apart—can be fully driven and programmed with light from a single optical fibre. Researchers used electrodes with a unique optical driver that creates programmable patterns of light via interference. This light travels through an optical fibre onto which nano lasers are printed. To demonstrate this

approach, the researchers used a high-resolution transfer-printing technique to fabricate multiple photonic crystal nano lasers spaced 18 microns apart. These arrays were applied onto the surface of a 2-micron-diameter optical microfibre. The interference pattern could also be modified by adjusting the driving beam’s polarisation and pulse width. The design allowed multiple nano laser arrays to be driven using light traveling through a single fibre.



Researchers have developed a new all-optical method for driving multiple high-density nanolaser arrays using light traveling down a single optical fibre. The optical driver creates programmable patterns of light via interference (Credit: Myung-Ki Kim, Korea University)

ML model to predict earthquakes and pandemics developed

Researchers at Brown University and MIT have developed a machine learning model that can predict an upcoming natural calamity like earthquakes and pandemics. They combined statistical algorithms—which need less data to make accurate, efficient predictions—with a powerful machine learning technique and trained the model to predict scenarios, probabilities, and sometimes even the timeline of rare events despite the lack of historical record on them. Researchers applied the pinpointing parameter and different ranges of probabilities for dangerous spikes during a pandemic, finding and predicting rogue waves, and estimating when a ship will crack in half due to stress. They found they could discover and quantify when rogue waves will form by looking at probable wave conditions that nonlinearly interact over time, leading to waves sometimes three times their original size.

Flaws in lithium-ion batteries being studied to improve them

An international group of experts, led by researchers from Texas A&M University, have combined powerful imaging techniques and large data sets to better understand why lithium-ion batteries fail and how they can be improved. Researchers were able to visualise the flaws and map defects